SYNTHETIC SAMPLE DATA: NOT AN OFFICIAL RECORD. Created for FDE AI Hackathon 5.0.

CSC 215: Data Structures in Java

Institution: Harborview Community College (fictional) **Credits:** 4 (semester hours) **Instructor:** Staff
Prerequisite: CSC 110 with C or higher **Contact hours:** 3 lecture + 2 lab hours per week, 15 weeks **Language:** Java 17

Course description

Second programming course focused on data structures and their analysis, taught in an object-oriented language (Java). Covers linked lists, stacks, queues, trees, binary search trees, heaps, hash tables and an introduction to graphs, with Big-O analysis of each operation.

Learning outcomes

- Implement linked lists, stacks, queues, binary search trees, heaps and hash tables in Java
- Analyse the time and space complexity of operations using Big-O notation
- Use interfaces and generics to build reusable collections
- Choose an appropriate data structure for a given problem

Weekly schedule

Week	Topic
1	Java review; classes and objects
2	Interfaces, generics and the Java Collections API
3	Algorithm analysis and Big-O notation
4	Arrays and dynamic arrays (ArrayList implementation)
5	Singly and doubly linked lists
6	Stacks: array and linked implementations; expression evaluation
7	Queues and deques; circular buffers
8	Recursion; midterm exam
9	Trees and tree traversals
10	Binary search trees: insert, delete, search
11	Heaps and priority queues; heapsort
12	Hash tables: hashing, chaining, open addressing, load factor
13	Sorting: merge sort, quicksort, comparison of algorithms
14	Introduction to graphs: representations, BFS and DFS
15	Final project presentations; final exam

Assessment

Weekly labs 25%, two programming projects 25%, midterm 20%, final exam 30%

Textbook: instructor-provided open course notes.

SYNTHETIC SAMPLE